CURRICULUM VITAE

✦ PERSONAL INFORMATION



Name: **Siyue YAO** Gender: Male
Major: Mechanical Engineering Education: B.S.
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✦ EDUCATION

2022.09 ~ Present Ph.D., School of Mechanical Engineering, Shanghai Jiaotong University
GPA: 3.95/4.0, Major Courses: Mechanism Theory, Mechanical Dynamics
Mentor: Prof. **Genliang Chen**

2018.09 ~ 2022.06 B.S., School of Mechanical Engineering, Shanghai Jiaotong University
GPA: 3.90/4.3, Major Courses: Robotics, System and Control

✦ RESEARCH INTERESTS

Compliant Mechanisms (end-effector) Design, Modeling, Sensing and Control for Grasping and Manipulation; Bio-inspired Robot Designs; Flexible Electronics; Soft Actuation

✦ MAIN ACADEMIC PUBLICATIONS

✧ Compliant End-effector with Force/Tactile Sensing:

[1] **S. Yao**, H. Yuan, C. Wu, Y. Wang, H. Wang, G. Chen. Modeling and Optimization of A Variable Stiffness Compliant Gripper with the Flexible Parallelogram Mechanism, ***IEEE/ASME Transactions on Mechatronics***, 2026.

[2] **S. Yao**, X. Xie, S. Zhao, Y. Li, H. Li, M. Cutkosky, G. Chen. Lego-Like Stiffness Configuration of Planar Compliant Modules for Task-Specific Flexible Interfaces, ***ICRA***, 2027. (Under Review)

[3] G. Chen, J. Luo, **S. Yao**, Y. Wang, Y. Xun, H. Wang, Z. Lin. Tactile sensing methodology based on propagation of contact-induced deformation in flexible beams, ***International Journal of Robotics Research***, 2024.

[4] **S. Yao**, T. Yan, Y. Wang, T. Liu, W. Li, Y. Zhao, H. Wang, G. Chen. Multi-Dimensional Force and Torque Sensing Along With Pose Estimation Based on a Compliant Fingertip With Sensor-Integrated Flexible Beam Arrays, ***ASME IDTEC/CIE***, Anaheim, California, USA, 2025.

[5] S. Tang, **S. Yao**, X. Zhu, T. Yu, H. Wang, G. Chen. Elastic Link Enables Structural Compliance and Intrinsic Contact Sensing in Robotic Legs, ***Mechanism and Machine Theory***, 2025.

✧ Bio-inspired and Soft Robotics:

[6] S. Yao, H. Li, Y. Lu, C. Wu, M. Cutkosky, G. Chen. EBUG: Elastic-beam Buckling and Undulating Generation for Multi-modal Robot Locomotion in Environmental Exploration, IROS, 2026.

[7] X. Xie, H. Li, S. Zhao, **S. Yao***, S. Song, M. Cutkosky. Fluorescence-enhanced Whisker Array with Vision-based Deformation Analysis for Underwater Source Localization, ***ICRA***, 2027. (Under Review)

[8] H. Li, T. Tu, **S. Yao**, Z. Chang, J. Jung, X. Xie, L. Chung, T. Ren, G. Chen, M. Cutkosky. Fiber Bragg Grating Whiskers for Bioinspired Hydrodynamic Perception on Underwater Robots, ***IEEE Transactions on Robotics***, 2026. (Under Review)

[9] J. Luo, **S. Yao**, T. Yan, T. Yu, H. Li, X. Duan, H. Wang, M. Cutkosky, G. Chen. A biomimetic multimodal whisker sensor for both contact and airflow sensing of Mobile robots, ***Sensors and Actuators A: Physical***, 2026.

[10] J. Luo, **S. Yao**, T. Yu, Y. Wang, H. Wang, G. Chen. Whisker-Like Compact Elastic Beam Sensor for Contact Position Estimation, ***IEEE Transactions on Industrial Electronics***, 2025.

[11] G. Chen, Y. Long, **S. Yao**, S. Tang, J. Luo, H. Wang, Z. Zhang, H. Jiang. A non-electrical pneumatic hybrid oscillator (PHO) for high-frequency multimodal robotic locomotion, ***Nature Communications***, 2024.

✧ **Modeling and Design of Traditional Mechanisms:**

[12] **S. Yao**, Y. Lu, X. Zhu, T. Yan, Y. Yang, H. Wang, G. Chen. On the kinematics of general plane-symmetric Bricard mechanisms. ***Mechanism and Machine Theory***, Vol. 190, pp. 105433, 2023.

[13] Y. Lu, **S. Yao**, Y. Wang, H. Wang, G. Chen. A Geometric Approach for the Bifurcation Modeling and Analysis of Plane-symmetric Bricard Mechanisms. ***6th IEEE/IFToMM International Conference on Reconfigurable Mechanisms and Robots***, Chicago, US, July 9-12, 2024.

✦ RESEARCH SPECIALTY

Analytical kinetostatic and stiffness modeling and co-design of flexible units and compliant mechanisms; FPC-based sensor and corresponding integrated circuit design; Force- / Network-based Control; System integration, including sensing, mechanism, and actuation; Fluid Simulation

✦ PROJECT EXPERIENCE (LEADING)

✧ **Compliant End-effector with Force/Tactile Sensing:**

2022.09 ~ 2025.02 Contact Sensing Based on Propagation of Contact-Induced Deformation
2022.09 ~ 2025.02 Stiffness-based Geometric Design for Flexible Parallel Gripper
2023.12 ~ 2024.12 Compliant Fingertip based on flexible beam array
2024.12 ~ 2025.12 Lego-like Modular Design for Stiffness Composition
2026.09 ~ Present A Sensing-Integrated Compliant Gripper for Force-aware Interaction

✧ **Bio-inspired and Soft Robotics:**

2022.09 ~ 2024.12 A Snap-through Triggered Pneumatic Hybrid Oscillator of High Frequency
2025.12 ~ Present Biomimetic Soft Pneumatically Actuated Fish
2024.12 ~ 2026.09 Strain Gauge- / FBG- / Vision-based Sensing for Whisker Array Designs
2026.09 ~ Present Actively Searching Whiskers for Object Localization and Tracking

✧ **Modeling and Design:**

2020.02 ~ 2021.02 Optimized Design of Minimally Invasive Surgical RCM Robot
2021.09 ~ 2022.06 Flexible Quadruped Walking Robot Design, Modeling and Analysis
2022.09 ~ 2023.06 Geometric Analysis Method for Plane-Symmetric Bricard Mechanisms

✦ AWARDS

2022.04 Challenge Cup Science and Technology Competition, **Star Prize**
2021.11 Shanghai Jiaotong University Rongchang Scientific Innovation Scholarship
2019.12 **National Scholarship** for Undergraduate and Junior College Students

✦ FUTURE WORK

(1) Investigate the properties of elastic units (damping) and achieve better dynamic performance
(2) Integrate bio-inspired hints about compliance into robot designs (motion, grasp, manipulation, etc.)
(3) Combine advantages of soft actuation, tactile sensing, compliant frame, etc. to build interesting robots